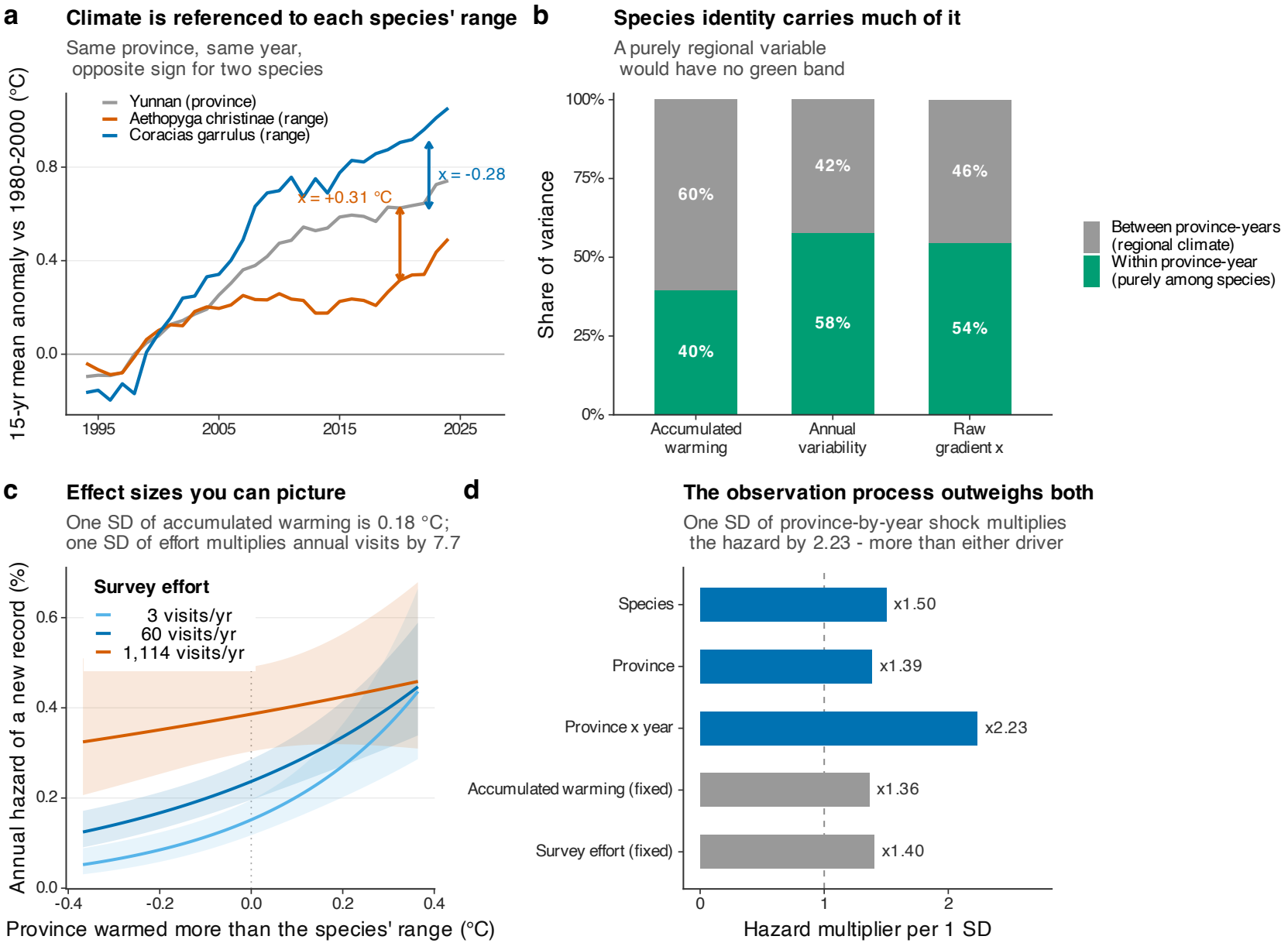
a
Climate is referenced to each species' range
Same province, same year,
opposite sign for two species
Yunnan (province)
Aethopyga christinae (range)
Coracias garrulus (range)
x = +0.31 °C
x = -0.28
15-yr mean anomaly vs 1980-2000 (°C)
0.8
0.4
0.0
1995
2005
2015
2025
b
Species identity carries much of it
A purely regional variable
would have no green band
Share of variance
100%
75%
50%
25%
0%
60%
40%
42%
58%
46%
54%
Accumulated warming
Annual variability
Raw gradient x
Between province-years (regional climate)
Within province-year (purely among species)
c
Effect sizes you can picture
One SD of accumulated warming is 0.18 °C;
one SD of effort multiplies annual visits by 7.7
Survey effort
3 visits/yr
60 visits/yr
1,114 visits/yr
Annual hazard of a new record (%)
0.6
0.4
0.2
0.0
-0.4
-0.2
0.0
0.2
0.4
Province warmed more than the species' range (°C)
d
The observation process outweighs both
One SD of province-by-year shock multiplies
the hazard by 2.23 - more than either driver
Species
Province
Province x year
Accumulated warming (fixed)
Survey effort (fixed)
x1.50
x1.39
x2.23
x1.36
x1.40
0
1
2
Hazard multiplier per 1 SD